- They are responsible for breaking down the nutrients and sugars that the cell receives and convert that into energy.
- It enables a plant to make ATP from a system in which the electrons are provided by chlorophyll that have been activated by light.

1.4.13.Vacuole:

The vacuole is the membrane-bound space found in the cytoplasm. Plant cells possess a well-developed vacuolar system, which becomes more prominent in maturing cells. It is also present in the cells of animals, fungi and bacteria but they are smaller in size. In plant cells the vacuoles can occupy up to 90 percent of the volume of the cell. Vacuoles contain water, sap, excretory product and other materials not useful for the cell. The vacuole is bound by a single membrane called tonoplast. In plants, the tonoplast facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole, hence, their concentration is significantly higher in the vacuole than in the cytoplasm. In Amoeba, the contractile vacuole is important for excretion. In many cells, as in protists, food vacuoles are formed by engulfing the food particles.

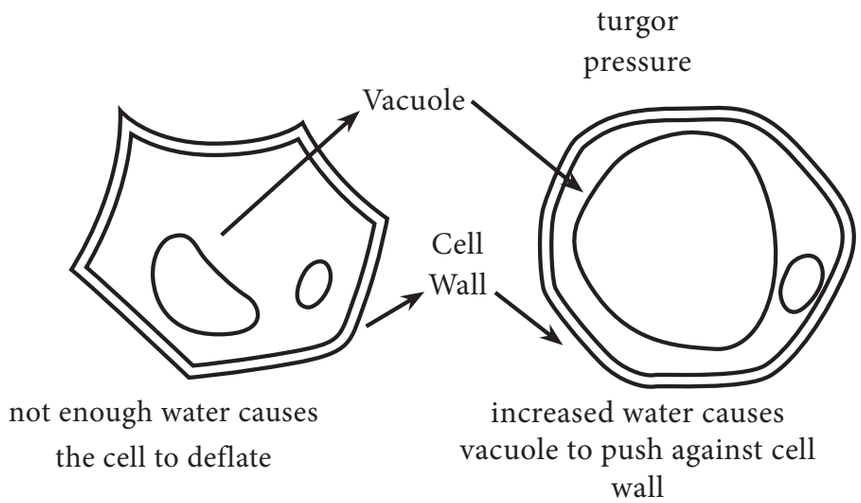


Figure 1.13 A schematic diagram of vacuole



In plants, inside each cell, there is a central vacuole, which can hold water, and under favourable conditions, the water enters the cell by osmosis (osmotic flow of water from area of low solute concentration outside the cell into the cell's vacuole, which has a higher solute concentration) and fills up the vacuole, creating a pressure called turgor pressure. This turgor pressure pushes the plasma membrane against the cell wall of plant, thereby stiffening the cell. This is mainly what makes non-woody parts of plants stiff and vertical.

In plant cells, the vacuoles accumulate a high concentration of sugars and other soluble compounds. Water enters the vacuole to dilute these sugars, generating hydrostatic pressure that is counterbalanced by the rigid wall. In this way the cells of the plant become stiff or turgid, in the same way that when an inner tube is inflated inside a bicycle tyre the combination

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